MILITARY SPECIFICATION

DETECTING-TRANSMITTING SET, ELECTROMAGNETIC, AN/GSQ-160( )

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Detecting-Transmitting Set, Electromagnetic, AN/GSQ-160( ). The AN/GSQ-160 is a portable intrusion detector capable of detecting intrusions within its detection range and transmitting an alarm to a remote monitoring station, while remaining unattended after hand emplacement.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification, to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-71	Pallet, Materials-Handling Wood (General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrguated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic (and connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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Military

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials and Processes Used In Electronic Equipment
MIL-M-13231	Marking for Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-B-55679	Battery, Dry (Special Purpose)

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40 inch x 48 inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods

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DRAWINGS

Electronic Command

SM-A-588552	Test Procedures, Burn-in for AN/GSQ-160
SM-A-588553	Test Procedure, Circuit Card Assy, Primary Processor
SM-A-588554	Test Procedure, Circuit and Assy, Secondary Processor
SM-A-588555	Test Procedure, Oscillator Assy, RF Sensor
SM-A-588556	Test Procedure, Coupler, Antenna Assembly
SM-A-588557	Test Procedure, Circuit Card Assy, Supplemental Circuits
SM-A-588558	Test Procedure, AS-2542/GSQ-160 Antenna Assy
SM-A-588559	Test Procedure, Detecting-Transmitting Set, MX-8846/GSQ-160
SM-A-599561	Test Procedure, Air Seal Test
SM-A-688563	Test Procedure, Quality Assurance Test, EMID III Group B (Appendix C)
SM-A-588564	Test Procedure, Quality Assurance Test, EMID III Group C (Appendix D)

DL's

DL-SM-B-588200	Detecting-Transmitting Set, AN/GSQ-160
DL-SM-B-588670	Detecting-Transmitting Set, MX-8846/GSQ-160
DL-SM-B-588672	Circuit Card Assy, Primary Processor (A3)
DL-SM-B-588673	Circuit Card Assy, Secondary Processor (A4)
DL-SM-B-588674	Circuit Card Assy, Supplemental Circuits (A2)
DL-SM-B-588675	Circuit Card Assy, Oscillator Assy, R.F. Sensor (A1)
DL-SM-B-588676	Coupler, Assy, Antenna (A5)
DL-SM-B-588679	Antenna AS-2532/GSQ-160

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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US Army Electronics Command

ECOM-0198-F18 — Reliability Acceptance Test Plan, AN/GSQ-160

Sandia Corporation

DS-P80515
DS-P62921
DS-P62526

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description. Detecting-Transmitting Set Electromagnetic, AN/GSQ-160 is a lightweight, self-powered portable intrusion device, designed for hand emplacement and unattended operation in forward combat zones by military personnel. The principle of detection employed is monitoring the target-induced frequency deviations of two r.f. oscillators, closely coupled to a single antenna. Upon detection, the transmitter is actuated and a message transmitted to a remote monitoring station. The detecting-transmitting set minimizes false alarms caused by all causes other than moving people, vehicles or boats. The AN/GSQ-160 performs its functions while remaining unattended after hand emplacement. The AN/GSQ-160 incorporates security functions that initiate a self-disable if unauthorized tampering occurs. The set is capable of detecting intrusions by walking personnel or vehicular traffic, when placed near a road or trail, or boat traffic, when emplaced on or near a body of water. The configuration is suitable for rapid burial of all components, except the day/night sensor, below natural grade level. The AN/GSQ-160, at the time of emplacement, may be set for either continuous operation or night only operation. The set is capable of carrying out successive missions for a period of two years with battery replacement only. The battery is capable of 45 days continuous operation. The set is capable of uninterrupted storage for five (5) years, and then performing all of its functions.

3.1.1 Equipment sub-systems. Detecting-Transmitting Set AN/GSQ-160 shall be comprised of the following major systems or modules:

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a. Antenna; AS-2542/GSQ-160

b. Sensor/Diplexer

c. Primary Processor

d. Secondary Processor

e. Power Supply; BA-1549(U) (GFP)

f. Supplemental Control Circuits, including:

(1) Day/Night Controls

(2) Arming Controls

(3) Disable/Controls

(4) Test Points

g. Data Transmission Modules (GFP)

(1) Transmitter, Radio; T-1233/GSQ (Commonly called TC-516)

(2) Encoder, Data Signals; Ky-678/GSQ (Commonly called TC-431)

(3) Code Plug; Ky-677/GSQ (Commonly called TC-432)

h. Chassis Assembly (Case)

3.2 First article samples. (4.3) The contractor shall furnish first article samples of Detecting-Transmitting Set AN/GSQ-160 for approval, as specified in the contract provisions.

3.3 Construction. Detecting-Transmitting Set AN/GSQ-160 shall be constructed in accordance with the drawing and data lists indicated below:

DL-SM-B-588200	Detecting-Transmitting Set, AN/GSQ-160
DL-SM-B-588670	Detecting-Transmitting Set, MX-8846/GSQ-160
DL-SM-B-588672	Circuit Card Assy, Primary Processor (A3)
DL-SM-B-588673	Circuit Card Assy, Secondary Processor (A4)
DL-SM-B-588674	Circuit Card Assy, Supplemental Circuits (A2)
DL-SM-B-588675	Circuit Card Assy, Oscillator Assy, R.F. Sensor (A1)

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DL-SM-B-588676 Coupler Assy, Antenna (A5)
DL-SM-B-588679 Antenna AS-2542/GSQ-160

3.4 Parts, materials, and processes. (4.4) Unless otherwise specified, electronic components, parts, devices, materials, techniques and technologies shall conform to MIL-P-11268.

3.4.1 Finish, protective. Equipment shall be given a protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary tough up after mounting. The color of the final paint film on type 1 surfaces shall be lusterless (olive drab) green conforming to #34087 of FED-STD-595.

3.4.2 Marking.

3.4.2.1 General. Marking shall conform to Specification MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.4.2.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disasssembly of the equipment.

3.4.2.3 Serial numbers. Detecting-Transmitting Set AN/GSQ-160 shall be serial numbered. (6.2).

3.4.2.4 Recovery Code. The recovery code shall be permanently marked, interior to the AN/GSQ-160. The location shall be on the underneath side of the top of the unit in that compartment where the data transmitter is located and only readable when the data transmitter is fully removed. A removable paper tag shall be attached to the AN/GSQ-160. The information contained on the tag shall be in accordance with Appendix B hereto.

3.4.3 Controls. All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.5 Preconditioning. (4.6) All units shall be bounce preconditioned in accordance with 3.5.1 prior to submission for electrical preconditioning in accordance with 3.5.2.

3.5.1 Bounce preconditioning. (4.6.1) All units shall meet the requirements herein, without subsequent processing, after subjection to the bounce pre-conditioning of 4.6.1 (also see 4.5).

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3.5.2 Electrical preconditioning. (4.6.2) All units shall meet the requirements herein, without subsequent processing, after subjection to the electrical preconditioning of 4.6.2.

3.6 Performance characteristics (4.7) Standard ambient conditions, whenever used herein, are defined as follows:

Temperature = 75° + 10°F
Relative Humidity = 50 + 30 percent
Atmospheric Pressure = 28.5 + 2.0; -4.5 in. of Hg.

3.6.1 System. (4.7.1)

3.6.1.1 Operating Voltages. The AN/GSQ-160 shall be capable of operation with a battery, BA-1549, having the following range of output voltages.

	Standard Ambient	Environmental
1. Positive low output voltage	4.16 to 3.2 VDC	4.16 to 3.40 VDC 4.16 to 3.45 VDC @ 33°F
2. Negative low output voltage	-4.16 to -3.2 VDC	-4.16 to -3.40 VDC -4.16 to -3.45 VDC @ 33°F
3. High output voltage	+30 to +20 VDC	+30 to +20 VDC

3.6.1.1.1 The current drain on the battery shall not exceed the following:

	Standard Ambient	Environmental
a. +4 VDC	9.5 ma, max	9.6 ma, max
b. -4 VDC	8.3 ma, max	8.3 ma, max
c. +6.3 VDC	1.1 ma, max	1.1 ma, max
d. +30 VDC	17 ua, max	17 ua, max

3.6.1.2 Operation, disable and control functions. The following devices shall be employed to control the self disable function of the data transmitter in accordance with Figure 1.

Arming/Recovery Switch
Code Select Switches
Tamper Switch
Day/Night Switch
Low Voltage Disable Circuit

3.6.1.2.1 Arming/Recovery Switch. The Arming/Recovery Switch shall be a 3 sequential position switch having the following positions: (a) Off; (b) ON/TEST; (c) Armed.

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3.6.1.2.2 Code Select Switches. Two twelve (12) position switches shall be provided, one with positions labelled Set, A, B, C through K, the other SET, 1, 2 through 11. A single Correct Code setting is designated for each AN/GSQ-160 unit. All other settings are not correct codes. A Correct Code setting may not include the SET position of either Code Select Switch.

3.6.1.2.3 Tamper Switch. The tamper switch is activated whenever the lower case of the AN/GSQ-160 is removed.

3.6.1.3 Day/Night Switch. The Day/Night Switch when active turns the AN/GSQ-160 "on" at night and "off" during day light. It is activated by circuitry containing a day/night photoelectric sensor according to the following specifications.

ON to OFF Switching Level - 4 lux to 16 lux
OFF to ON Switching Level - .15 to .25 of level required to switch that unit from ON to OFF.

3.6.1.4 Low Voltage Disable Circuit. This circuit shall be activated in accordance with Figure 1 and whenever any of the battery levels fall below the specified levels.

Voltage	Disable Level
+ 4.0 volt	+ 3.4 ± 0.2
- 4.0 volt	- 3.4 ± 0.2
+ 30 volt	+ 21 ± 1

Figure 1 - Self Disable Matrix

Arming Recovery Switch	OFF		ON/TEST		ARMED	
Day Night Switch	OFF(Day)	ON(Night)	OFF(Day)	ON(Night)	OFF(Day)	ON(Night)
Activate Tamper Switch	N	N	N	N	D	D
Low Voltage (30, +4 or -4)	N	N	N	N	N*	D
Change Arming/Recovery Switch while Correct Code is set	N	N	N	N	N	N
Change Arming/Recovery Switch while Correct Code is not set	N	N	N	N	D	D

* Disable will occur when Day Night Switch switches to ON (Night).
N = No disable D = Disable

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3.6.1.5 Sensor frequencies. With the AN/GSQ-160 in the "inside environment" configuration, specified in 3.6.1.6, the sensor frequencies shall be as follows:

	Standard Ambient	Environment
a. High frequency	59.850 to 60.150 MHZ	59.800 to 60.200 MHZ
b. Low frequency	57.350 to 57.650 MHZ	57.300 to 57.700 MHZ
c. Difference frequency	2.440 to 2.560 MHZ	2.400 to 2.600 MHZ
d. Power - Total RF	2.0 mw, max	2.0 mw, max

3.6.1.6 Close target rejection. The frequency deviation response of each oscillator is a sinusoidal function whose amplitude and phase characteristics are both functions of a target's range. The sensitivity of the AN/GSQ-160 to close-in targets is then completely determined by the amplitude and phase characteristics of each oscillator's sinusoidal deviation function. Within an inside environment, the only means available to determine the above sinusoidal characteristics, is by simulating both the antenna and target with an Antenna Simulator (SM-D-588207), and Target Simulator (SM-D-588209). Below is specified a dual listing of parameters for (1) the AN/GSQ-160 as fully deployed outside and (2) the AN/GSQ-160 as deployed within an inside environment, described above.

$$R(r) = \frac{[\text{Average of the frequency deviation response of each oscillator at range, r}]}{[\text{Difference frequency deviation response at range, r.}]}$$

$$A = \frac{[\text{Difference frequency deviation @ equivalent range of 7.0' - Difference frequency deviation @ equivalent range of 2.75'}]}{[\text{Low frequency deviation at an equivalent range of 5.0'}]}$$

$$D = \frac{[\text{Difference frequency deviation @ 4.75'}]}{[\text{High frequency deviation @ 5.0' - Low frequency deviation @ 5.0'}]}$$

OUTSIDE	INSIDE
a. The range closest to the antenna, for which the difference frequency deviation response is zero shall be equal to 5.2 ± 1 ft.	The equivalent range, simulated through the switching matrix, for which the high frequency oscillators deviation frequency response reverses polarity, shall be between 2.5 and 3.0 ft.

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b. The range, closest to the antenna, for which the sinusoidal frequency deviation functions, for both oscillators, are in phase shall be 8 ± 1 ft.	The function A, defined above, shall be equal to 0.12 ± 0.05.
c. The magnitude R, defined above, at the range specified in b, shall be NLT 20.	The function D, defined above, shall be equal to 0.04 ± 0.02.

3.6.1.7 Difference frequency deviation sensitivity. With the AN/GSQ-160 in the "inside environment" configuration, the difference frequency deviation sensitivity is defined as the incremental change in sensor difference frequency produced by the switching matrix (SM-D-588368) at a range of 29 feet.

Standard Ambient	Environment
230 - 330 hz	220 - 350 hz

3.6.1.8 Data transmitter. The AN/GSQ-160 shall incorporate a data transmitter comprised of the modules listed below. The data transmitter shall provide a coded r.f. alarm transmission coincident with and as a result of the detection of a target. The data transmitter may or may not be furnished to the contractor for incorporation within the delivered equipment. (6.2.c). Nevertheless, the contractor shall be fully responsible for the AN/GSQ-160 being an operable system data transmitter comprised of the modules and their specifications listed below:

Item	Specification
Transmitter, Radio; T-1233/GSQ	DS-P80515 (Sandia Corp)
Encoder, Data Signals; KY-678/GSQ	DS-P62921 (Sandia Corp)
Code Plug; KY-677/GSQ	DS-P62526 (Sandia Corp)

3.6.1.9 Mechanically induced antenna noise. (4.7.1.1) With the antenna deviated from its unperturbed state, a maximum of 2.8 inches at the tip, in any direction, the primary processor output shall not have as a voltage output, a signal equivalent to 200 hz of difference frequency deviation, max.

3.6.2 Primary Processor. (4.7.2). The Primary Processor of the AN/GSQ-160 consists of limiting amplifiers feeding a Weiss type discriminator. The discriminator is followed by an Amplifier/Filter having a 4 khz window centered about the mean frequency of the signal input. The function of the

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Primary Processor is to convert the frequency variations of the signal input to an analog voltage.

3.6.2.1 Adjustments. There are two (2) adjustments within the Primary Processor. These adjustments allow for discriminator tuning at 2.700 MHz and 2.300 MHz. The adjustment procedures are described in SM-A-588553.

3.6.2.2 Characteristics.

		Standard Ambient	Environmental
3.6.2.2.1	Scale factor @ 2.5 MHz	1.08 to 1.32 mv/hz	1.0 to 1.14 mv/hz
3.6.2.2.2	Scale factor @ 2.38 MHz	0.96 to 1.32 mv/hz	0.8 to 1.4 mv/hz
3.6.2.2.3	Scale factor @ 2.62 MHz	0.96 to 1.32 mv/hz	0.8 to 1.4 mv/hz
3.6.2.2.4	Slewing Time; 2.50 to 2.62 MHz	45 to 75 sec.	45 to 85 sec.
3.6.2.2.5	Slewing time; 2.62 to 2.38 MHz	95 to 155 sec.	115 to 175 @ 165°F 95 to 155 @ 33°F
3.6.2.2.6	AM rejection @ 2.38 MHz	100 mv, max	100 mv, max
3.6.2.2.7	AM rejection @ 2.62 MHz	100 mv, max	100 mv, max
3.6.2.2.8	Output Noise with Stable Input	1.5 mv, arv, max	1.9 mv, arv, max
3.6.2.2.9	Output Bias	0.9 to 1.3 VDC	0.6 to 1.9 VDC
3.6.2.2.10	Filter Response to Step Function		
	(1) Rise Time to 63%	0.068 to 0.092 sec	0.064 to 0.096 sec
	(2) Fall Time to 37%	3.2 to 4.8 sec	3.0 to 5.0 sec
3.6.2.2.11	Output, with muting		
	(1) DC level	0.85 to 1.15 VDC	0.80 to 1.20 VDC
	(2) AC Variation	50 mv, max	100 mv, max
	(3) Muting Period	80 to 120 msec	70 to 130 msec

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3.6.2.2.12 Operating Currents

(1) + 4 VDC @ 2.5 MHz	1.5 to 2.0 ma	1.4 to 2.1 ma
(2) - 4 VDC @ 2.5 MHz	1.8 to 2.6 ma	1.6 to 2.8 ma
(3) + 2.8 VDC @ 2.5 MHz	0.05 to 0.09 ma	0.04 to 0.1 ma
(4) - 2.8 VDC @ 2.5 MHz	1.4 to 2.1 ma	1.3 to 2.2 ma

3.6.3 Secondary processor. (4.7.3) The secondary processor discriminates intruder signatures from background noise to detect the presence of targets. When after having determined that a target is present, it then generates a trigger signal to the data transmitter. The secondary processor contains circuitry to perform the following functions:

(1) Logarithmic amplification.

(2) Linear, full-wave rectification.

(3) Band-pass filtering.

(4) Pulse generation to the data transmitter whenever the output of the bandpass filter exceeds a threshold.

(5) The generation of a muting pulse during the period of a data transmitter activation.

3.6.3.1 Adjustments. The single adjustment required in the secondary processor is that of the gain in the integrated circuit amplifier designated as Z3 and controlled by the variable resistor R39. The adjustment procedures are described in SM-A-588554.

3.6.3.2 Characteristics

	Standard Ambient	Environment
3.6.3.2.1 Initial turn-on voltage (Pin E)	-2 VDC, max	-2 VDC, max
3.6.3.2.2 Settling Time (Pin E)	7 min, max	7 min, max
3.6.3.2.3 Settled Voltage (Pin E)	-1.7 to -1.9 VDC	-2.0 to -2.2 VDC @ 165°F -1.58 to -1.78 @ 33°F

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3.6.3.2.4 Alarm Threshold (above noise)	10.0 to 12.0 db	9.5 to 12. 5 db
3.6.3.2.5 Log Compressor Bias	-0.1 to +0.1 VDC	-0.2 to +0.2 VDC
3.6.3.2.6 Log Compressor Output:		
(1) The average of the magnitude of the increase in positive peak output voltage and the magnitude of the increase in negative peak output voltage for an increase of input voltage from 4 mv p-p to 8 mv p-p. (1 HZ square wave input).	32 to 40 mv	27 to 39 mv @ 33°F 36 to 48 mv @ 165°F
(2) The average of the magnitude of the increase in positive peak output voltage and the magnitude of the increase in negative peak output voltage for an increase of input voltage from 400 mv p-p to 800 mv p-p. (1 HZ square wave input).	32 to 40 mv	27 to 39 mv @ 33°F 36 to 48 mv @ 165°F
(3) Constant Difference - The difference between the average output increase in (1) and the average output increase in (2) for both positive and negative peaks.	± 2 mv	± 3 mv
(4) Symmetry - The difference between the magnitude of the output positive peak increase and the output negative peak increase in (1) and in (2).	± 6 mv	± 6 mv
(5) Level, during mute	-0.1 to +0.1 VDC	-0.2 to +0.2 VDC
(6) Period for Step 5	2.8 to 4.2 sec	2.5 to 4.5 sec
3.6.3.2.7 Secondary Processor Output		
(1) Rise Time (0 to 63%)	13 to 17 sec	11 to 19 sec
(2) Fall Time (90 to 33%)	63 to 79 sec @ 33°F @ 165°F	56 to 86 sec 72 to 120 sec
(3) Alarm Pulse	5.9 to 6.3 VDC	5.9 to 6.3 VDC
(4) Alarm Period	35 to 45 msec	35 to 45 msec

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3.6.3.2.8	Repetitive Alarms		
	(1) Time interval between 1st and 2nd alarm	20 to 30 sec	20 to 30 sec
	(2) Time interval between 2nd and 3rd alarm or	26.4 to 39.6 sec	26.4 to 39.6 sec
	2nd to 90% of 3rd buildup if no third alarm	21 to 33 sec	21 to 33 sec
3.6.3.2.9	Mute Output		
	(1) Normal (before mute)	-2.1 to -4.1 VDC	-2.1 to -4.1 VDC
	(2) Mute level	3.6 to 4.0 VDC	3.5 to 4.1 VDC
	(3) Mute Period (50% Point)	33 to 47 msec	30 to 50 msec
3.6.3.2.10	De-Logarithmic Compression		
	(1) +9.5 db, step (Defined as "E")	1.55 to 1.65 VDC	1.50 to 1.70 VDC
	(2) +6 db, step	(.5 to .6) x E	(.45 to .65) x E
	(3) +11 db, step	(1.15 to 1.25) x E	(1.1 to 1.3) x E
3.6.3.2.11	Per Cycle Limit	0.43 to 0.57 v per 1/2 cycle	0.35 to 0.65 v per 1/2 cycle
3.6.3.2.12	Noise Memory	-0.25 to +0.75 VDC	-0.5 to +1.0 VDC
3.6.3.2.13	Operating Currents		
	(1) + 4 VDC	0.7 to 1.1 ma	0.6 to 1.25 ma
	(2) - 4 VDC	0.75 to 1.2 ma	0.65 to 1.35 ma
	(3) + 2.8 VDC	0.4 to 0.6 ma	0.35 to 0.7 ma
	(4) -2.8 VDC	0.4 to 0.6 ma	0.35 to 0.7 ma
	(5) +6.3 VDC (during alarm)	0.35 ma, max	0.35 ma, max
	(6) +6.3 VDC (steady state)	0.0 to 10 ua	0.0 to 10 ua

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3.6.3.3 Secondary processor functional test. It shall be required that the secondary processor successfully pass the test of 4.7.3.1.

3.6.4 Sensor/oscillator. (4.7.4) The sensor/oscillator consists of two oscillators operating at a nominal frequency difference of 2.5 MHz, a mixer for extracting the 2.5 MHz difference signal and the circuitry to couple the two oscillators to a single output connector.

3.6.4.1 Adjustments. There are four (4) adjustments required to the sensor/oscillator. They are:

(1) Hi oscillator frequency
(2) Lo oscillator frequency
(3) Phase adjustment capacitor
(4) Balance potentiometer to adjust balance deviation sensitivity.

The adjustment procedures are described in SM-A-588555.

3.6.4.2 Characteristics

	Standard Ambient	Environmental
3.6.4.2.1 Current @ +2.8 VDC	5.4 ma, max	5.4 ma, max
3.6.4.2.2 RF Power; High frequency	1.0 mw, max	1.0 mw, max
3.6.4.2.3 RF Power; Low frequency	1.0 mw, max	1.0 mw, max
3.6.4.2.4 Difference frequency, amplitude	13 mv, rms, min	10.5 mv, rms, min
3.6.4.2.5 CTRR. CTRR, as measured at the sensor/oscillator, is determined through the measurements specified below.		
3.6.4.2.5.1 High frequency spacial phase crossover. The indicated range for the high frequency, spacial phase, polarity reversal.	2.25' - 2.75'	2.25' - 2.75'
3.6.4.2.5.1.1 Alignment of the high frequency spacial phase crossover. Initial alignment of this parameter, at standard ambient conditions.	2.35' - 2.75'	N/A

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3.6.4.2.5.2 "A" (figure of merit for the in-phase range of the sensor/oscillator).	0.016 ± 0.032	0.016 ± 0.032
3.6.4.2.5.3 D_{min}.	0.035 ± 0.008	0.035 ± 0.008
3.6.4.2.6 Sensor Oscillator FM Noise	4.2 Hz, rms, max	4.7 Hz, rms, max
3.6.4.2.7 Frequency		
(1) High frequency	60.050 to 59.980 MHz	60.165 to 59.865 MHz
(2) Low frequency	57.550 to 57.480 MHz	57.665 to 57.365 MHz
(3) Difference frequency	2.475 to 2.525 MHz	2.450 to 2.550 MHz @ 175°F 2.470 to 2.530 MHz @ 25°F

3.6.5 Supplemental Circuits. (4.7.5) The supplemental circuits perform the following function:

(1) Provide a regulated plus and minus 2.8 VDC from the ± 4 VDC battery.

(2) Turn the equipment on and off in response to the stimulus of the photo cell contained in the housing sub-assembly or the arming/recovery switch.

(3) Contains the circuitry for implementing the disable mechanism whenever the following shall occur:

(a) Lower case is removed.

(b) The Arming/Recovery Switch is moved from the "armed" position without having the code select switches set to the proper code.

(c) Battery voltages fall below:

+3.45 ± .01 VDC
-3.45 ∓ .01 VDC
+21.0 ∓ 1.0 VDC

3.6.5.1 Adjustments. None.

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6.5.2 Characteristics

		Standard Ambient	Environmental
3.6.5.2.1	+2.8 VDC load regulation; 2.5 to 8.0 ma load current	2.76 to 2.84 VDC	2.70 to 2.93 VDC
3.6.5.2.2	+2.8 VDC line regulation; 3.4 VDC (3.45 VDC @ +33°F) to 4.16 VDC	± 10 mVDC	± 15 mVDC
3.6.5.2.3	-2.8 VDC load regulation 2.5 to 8.0 ma load current	-2.76 to -2.84 VDC	-2.70 to -2.93 VDC
3.6.5.2.4	-2.8 VDC line regulation; -3.4 (-3.45 VDC @ +33°F) to -4.16 VDC	± 10 mVDC	± 15 mVDC
3.6.5.2.5	Noise Voltage	1 mv, peak-to-peak	2 mv, peak-to-peak
3.6.5.2.6	Day/Night Switch Level		
	(1) Night to Day Switching Level	100 to 135 kohms	100 to 135 kohms
	(2) Day to Night Switching Level Ratio of (1) above	420 to 590 kohms	420 to 590 kohms
	(3) On State; Expressed as the voltage of P1-F relative to P1-C.	+0.0 VDC -0.1 VDC	+0.0 VDC -0.2 VDC
	(4) On State; Expressed as the voltage of P1-J relative to P1-K.	+0.1 VDC -0.0 VDC	+0.2 VDC -0.0 VDC
	(5) Off State; Expressed as the leakage current between P1-F and P1-J respectively, to ground.	1 ua, max	10 ua, max
3.6.5.2.7	Disable Pulse		
	(1) Amplitude (into a nominal 7 ohm load)	2.3 VDC, min	2.1 VDC, min
	(2) Pulse period	10-26 msec	7-29 msec

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(3)	Rise Time	8 usec, max	10 usec, max
(4)	Fall Time	8 usec, max	10 usec, max

3.6.5.2.8 Low Voltage Disable Point

(1)	+ 4 VDC	3.3 to 3.5 VDC	3.2 to 3.6 VDC
(2)	- 4 VDC	-3.3 to -3.5 VDC	-3.2 to -3.6 VDC
(3)	+30 VDC	20.5 to 21.5 VDC	20 to 22 VDC

3.6.5.2.9 Input Current; Armed Condition

(1)	+ 4 VDC into nominal 1210 ohm resitive load	8.7 to 9.1 ma	8.6 to 9.2 ma
(2)	- 4 VDC into a nominal 976 ohm resitive load	7.9 to 8.3 ma	7.8 to 8.4 ma
(3)	+6.3 VDC	0.8 to 1.1 ma	0.7 to 1.2 ma
(4)	+30 VDC	13 to 17 ua	11 to 25 ua

3.6.6 Antenna. (4.7.6) The antenna shall radiate both the sensor and data transmitter frequencies. The antenna shall consist of two sections, connected together by a screw thread to form a single entity. In this configuration the upper section of the antenna shall function as an end-fed, halfwave dipole for the data transmitter while the entire antenna serves as a quarter-wave antenna at the sensor frequencies. The lower section of the antenna shall incorporate the necessary configuration to accomplish this.

3.6.6.1 Characteristics

		Standard Ambient	Environmental
3.6.6.1.1	VSWR between 162-174 MHz	2.5:1, max	2.5:1, max
3.6.6.1.2	Length		
(1)	Upper section	33 inches	
(2)	Lower section	15 inches	

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(3)	Curvature	6 ± 1 inches deviation of tip from a straight line.
(4)	Counterpoise	62 ± 1 inches

3.6.7 Antenna Coupler. The antenna coupler permits the data transmitter and the sensor/diplexer to couple to the antenna while minimizing the loading effect of each on the other. The antenna coupler consists of a length of coaxial cable, approximately 36 inches in length, from the data transmitter to the antenna, would to an inductance of 0.22 uhy, a coaxial cable from the sensor/diplexer to the antenna (the center conductor of which is tied to the shield of the data transmitter coaxial cable), a fixed capacitor, a fixed inductor and a variable capacitor. The wound coaxial cable and the three passive elements, together with any stray capacitance, form a parallel resonant circuit that isolates the sensor/oscillator from the data transmitter. The circuit phase shift characteristics of this circuit effect the spacial phase characteristics of the electromagnetic field established by the radiation of the two oscillator frequencies.

3.6.7.1 Adjustments. The sole adjustment required is to time the variable capacitor, C1, to obtain the input impedance of 3.6.7.2.2 and 3.6.7.2.3. This adjustment procedure is described in SM-A-588556.

3.6.7.2 Antenna coupler characteristics

	Standard Ambient	Environmental
3.6.7.2.1 Loss in cable to data transmitter between 162 to 174 MHz	0.8 db, max	0.8 db, max
3.6.7.2.2 Magnitude of J3 input impedance @ 100 MHz	33 ± 10 ohms	33 ± 10 ohms
3.6.7.2.3 Phase angle of J3 input impedance @ 100 MHz	82 ± 3 degrees	82 ± 3 degrees

3.7 Service conditions. (4.8) The detecting-transmitting set shall be capable of meeting the following service conditions.

3.7.1 Temperature. (4.8.1)

3.7.1.1 Low temperature. (4.8.1.1) The equipment shall operate to specified performance at ambient temperatures as low as +33°F, and shall withstand prolonged storage at -70°F.

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3.7.1.2 High temperature. (4.8.1.2) The equipment shall operate at specified performance at ambient temperatures as high as +113°F plus the effects of solar radiation at a rate of 360 BTU/ft^2/hr, and shall withstand storage at +165°F.

3.7.2 Relative humidity. (4.8.2) The equipment shall withstand exposure to high relative humidity of 94 ± 4% at 86°F.

3.7.3 Altitude. (4.8.3) The equipment shall operate without degradation in specified performance altitudes to 10,000 feet above seal level, and shall withstand air transport at altitudes to 50,000 feet.

3.7.4 Salt fog. (4.8.4) The equipment shall be capable of operating after being in a salt fog environment.

3.7.5 Sand and dust. (4.8.5) The equipment shall be capable of withstanding the effects of dry dust (fine sand) laden atmosphere.

3.7.6 Vibration. (4.8.6) The equipment shall have no resonances in the frequency range of 10 to 55 Hz that exceed twice the amplitude of the applied variation.

3.7.7 Bounce; Loose cargo. (4.8.7) The equipment shall be capable of withstanding shock encountered in transport.

3.7.8 Shock, Bench Handling. The equipment shall be capable of withstanding shock encountered in servicing.

3.7.9 Shock; drop. (4.8.9) The equipment shall be capable of withstanding drops on each face, edge, and corner (a total of 26 drops) from a height of 48 inches.

3.7.10 Immersion. (4.8.10) The equipment shall withstand submergence in 15 feet of seawater, while operating, due to two cycles of tides. The equipment shall meet performance requirements within three minutes after wave motion no longer impinges on the case.

3.7.11 Fungus. (4.8.11) Operation of the equipment shall not be impaired by exposure to fungus growtn as encountered in tropical climates.

3.7.12 Wind. (4.8.12) The equipment shall be capable of operating in winds at 50 mph with gusts to 70 mph.

3.8 Workmanship. (4.9) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268.

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3.9 Interchangeability and replaceability. (4.10) Parts, components, and assemblies shall be functionally, electrically and mechanically interchangeable at and above module level without resorting to selective assembly, Interchangeability for electronic equipment shall be in accordance with MIL-STD-454, Requirement 7. Below the module level, components may be selected during the manufacturing process so that completed modules meet performance requirements.

3.10 Air seal test. (4.11) The decrease in vacuum during the 100 second period shall not exceed .17 inches of water when tested with battery and data transmitter removed.

3.11 Electromagnetic interference. (4.12) The AN/GSQ-160 shall meet the RS-03 requirements of MIL-STD-461 for class Ia and Ib equipment. The set susceptibility shall be defined as a change of 20 Hz in primary processor output over the range of frequencies required by RS-03, but excluding the band from 25 MHz to 125 MHz. The degradation within the band 25 MHz to 125 MHz shall be minimized. Any other degradation of equipment performance, not directly attributable to changes in sensor oscillator frequency, are not allowable at any frequency and power level within the RS-03 test limits.

3.12 Reliability. The specified reliability shall be 92.2% probability of success for a 1080-hour mission under the following operating conditions:

a. Ambient operating temperature: 95°F to 113°F
b. Ground environment
c. 95% + 5% Relative humidity
d. Continous operation

This requirement does not include the T1143/GSQ, KY 678/GSQ or KY-677/GSQ but does include the interfaces between them and the AN/GSQ-160.

3.12.1 Reliability program. The contractor shall provide for, but not be limited to, the reliability tasks described in MIL-STD-785:

a. Reliability Organization (5.1.1)
b. Management and Control (5.1.2)
c. Subcontractor and supplies reliability programs (5.1.3)
d. Qualification testing (5.3.2)
e. Integrating equipment (5.1.2.2)
f. Defective or inadequate parts/specification (5.2.3)
g. Critical items (5.2.5)
h. Effects of storage, shelf-life, packaging, transportation, handling and maintenance (5.2.6)
i. Failure data collection, analysis, and corrective action (5.4.1)
j. Reliability demonstration (5.3.3)

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3.13 Technical literature. Technical literature shall be furnished as specified in the contract. (6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements specified herein. The contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein. All facilities must be acceptable to the Government.

4.1.1 Contractor product assurance. The contractor shall provide and maintain a means of determining product conformance in accordance with the requirements specified herein or elsewhere in the contract. The Government, at its option, may perform any evaluation deemed necessary to assure the adequacy of the means employed and the effectiveness of the contractor's product assurance methods and systems.

4.1.2 Government verification. All quality assurance operations required of the contractor shall be subject to Government verification at scheduled or unscheduled intervals. Verification will consist of the following:

a. Surveillance of the contractor's operations to determine that practices, methods, and procedures of the contractor's product assurance system are being applied and are, in fact, accomplishing the objectives of total compliance of the product with the requirements of this specification.

b. Government product inspections to determine compliance of the product with the specification requirements.

4.1.3 Accommodations and assistance. The Government quality assurance representative shall have the right to access to any area of the contractor's or his subcontractor's premises where any part of the work is being performed. The Government quality assurance representative shall be afforded unrestricted opportunity to verify conformance of the product with specification requirements. The contractor shall make his inspection equipment available for use by the Government quality assurance representative for verification purposes. The contractor's personnel shall be made available for operation of such inspection equipment as required.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection. (Does not include preparation for delivery.) (See 4.3.)

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b. Inspections covered by subsidiary documents. (See 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspections of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.14).

4.3 First article inspection. (3.2) This inspection shall be conducted on samples which have been assembled using parts, materials, and processes which will be employed in production, and shall be fabricated and assembled by methods similar to those used in production. This inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4. and the inspections shown in Table 1.

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TABLE I - 1st ARTICLE TESTING

Inspection In the order shown (Note 1)	Category Number 1	2	3	Req. Para.	Insp. Para.
In-process tests		all		Table II	
Group A Tests		all		Table III	
Group B Tests		all		Table IV	
Group C Tests				Table V	
Temperature, low	1				
Temperature, high	2				
Shock-Bounce		3			
Shock-Drop		4			
Immersion		5			
Humidity			1		
Other Tests					
Altitude	3			3.7.3	4.8.3
Wind	5			3.7.12	4.8.12
Sand and dust	6			3.7.5	4.8.5
Salt fog (Note 2)	7			3.7.4	4.8.4
Vibration		1		3.7.6	4.8.6
Shock Bench Handling		2		3.7.8	4.8.8
EMI	4			3.11	4.12
Fungus (Note 2)		6		3.7.11	4.8.11
Order for Inspection - 1,2,					

NOTE 1: Other first article inspection may precede, follow, or be interspersed between the non damaging tests.

NOTE 2: The equipment shall be thoroughly washed, cleaned, dried and refurbished, if necessary, before proceeding with subsequent tests.

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4.4 Inspection covered by subsidiary documents. (3.4). The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.

Item	Where Required
Finish	3.4.1
Marking	3.4.2
Workmanship	3.8

4.5 Quality conformance inspection of equipment before preparation for delivery. Every equipment shall be subjected to preconditioning (3.5 & 4.6) prior to conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records as an element of Government verification (see 4.1.2).

4.5.1 Group A Inspection. Each unit of equipment manufactured shall be inspected for conformance to all the examinations and test requirements in Table II. Discrete lots shall then be formed from units conforming to all the above tests, utilizing the procedures of MIL-STD-105 for lot formation. Quality verification of each discrete lot shall be performed utilizing the procedures of MIL-STD-105, using general inspection levels and the acceptable quality level (AQL) indicated in Table II. Lot size shall be 51-90 (sample size 13). Group A lot inspection shall be performed in any order which is satisfactory to the Government, except that the air seal test (4.11) shall be last. No adjustment shall be permitted during Group A verification.

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TABLE II - GROUP A INSPECTION

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor	Insp Level
1. Visual and Mechanical	3.8	4.9	1.0%	4.0%	I
Primary Processor	3.8	4.9	1.0%	4.0%	
Secondary Processor	3.8	4.9	1.0%	4.0%	
Sensor/diplexer	3.8	4.9	1.0%	4.0%	
Supplemental circuits	3.8	4.9	1.0%	4.0%	
Antenna	3.8	4.9	1.0%	4.0%	
Antenna coupler	3.8	4.9	1.0%	4.0%	
Electrical assembly	3.8	4.9	1.0%	4.0%	
Housing assembly	3.8	4.9	1.0%	4.0%	
2. Module *					
Primary Processor	3.6.2	4.7.2	1.0%		
Secondary Processor	3.6.3	4.7.3	1.0%		
Sensor/oscillator	3.6.4	4.7.4	1.0%		
Supplemental circuits	3.6.5	4.7.5	1.0%		
Antenna	3.6.6	4.7.6	1.0%		
Antenna coupler	3.6.7	4.7.7	1.0%		
3. System *	3.6.1	4.7.1 (except 4.7.1.1)	1.0%		
4. Air seal test *	3.10	4.11	1.0%		

* All failures are deemed to be major.

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4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table III and to procedures for general inspection levels of MIL-STD-105. Group B inspection shall be performed on production lots of the product that have passed Group A inspection. (See 4.5.2.1 for lot formation.) The inspections listed in Table III shall be conducted on those modules disassembled from the lot of equipment selected for Group B testing. As a part of the Group B tests, these modules will, at the conclusion of the module level testing, be randomly reassembled into complete units and reinspected under the system level Group A inspection procedures. This random reassembly and successful passing of the system level Group A inspection shall constitute the passing the interchangeability demonstration under these Group B inspection requirements. Test shall be conducted in accordance with SM-A-588563.

4.5.2.1 Group B sampling plans. The group B sampling plan, for the AQL's listed below, shall be as follows:

AQL	Inspection Level
2.5%	I (except that lot size shall be between 51 and 90 units. Sample Size 5)

TABLE III - GROUP B INSPECTION

Inspection	Requirement Paragraph	Inspection Paragraph (Note 1)
Primary processor	3.6.2	4.7.2
Secondary processor	3.6.3	4.7.3
Sensor/oscillator	3.6.4	4.7.4
Supplemental circuits	3.6.5	4.7.5

NOTE 1: Except test procedure SM-A-588563 shall be utilized in lieu of the procedure listed in the inspection paragraph.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C inspection. This inspection shall consist of the tests specified in Table IV and shall be performed on sample units that have been subjected to and met Group A an/ Group B inspection.

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TABLE IV - GROUP C INSPECTION

Inspection	Category	Sample Size	Requirement Paragraph	Insp. Para. Drawing
Temperature, low	1	1	3.7.1.1	4.8.1.1
Temperature, high	1	1	3.7.1.2	4.8.1.2
Bounce, Loose Cargo	2	1	3.7.7	4.8.7
Immersion	2	1	3.7.10	4.8.10
Humidity	3	1	3.7.2	4.8.2
Drop	2	1	3.7.9	4.8.9
Interchangeability	All units selected for Group C Category 1, 2 & 3		3.9	4.10

4.5.3.1 Sampling for Group C inspection. Three (3) sample units shall be selected at random for all inspections in Table IV from every 500 units and fraction thereof produced. One (1) unit each shall be assigned to category of test 1, 2 and 3. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.4 Non-compliance. The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the quality assurance representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the government) and group A and B inspections may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.5.5 Reinspection of conforming Group B and Group C sample units. Unless otherwise specified sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Preconditioning . (3.5)

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4.6.1 Bounce preconditioning. (3.5.1). The AN/GSQ-160 shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch thick strip of material under one corner or edge of the equipment ot a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this precondition for one minute. The antenna shall not be attached but shall be placed separately on the package tester during the test.

4.6.2 Electrical preconditioning. (3.5.2). All detecting-transmitting sets on order, when tested according to the production burn-in plan (debugging) SM-A-588552 for the equipment, shall meet the requirements contained therein prior to submission to acceptance inspection.

4.7 Performance characteristics. (3.6)

4.7.1 System. (3.6.1). The following measurements shall be made in accordance with SM-A-588559:

Current drain (3.6.1.1)
Arming and recovery switches (3.6.1.2.1)
Code Selection Switches (3.6.1.2.2)
Tamper Switch (3.6.1.2.3)
Day/Night Sensor (3.6.1.3)
Low Voltage Disable Circuit (3.6.1.4)
Sensor frequencies (3.6.1.5)
Close Target Rejection (3.6.1.6)
Difference frequency deviation sensitivity (3.6.1.7)
Data Transmitter (3.6.1.8)

4.7.1.1 Mechanically Induced Antenna Noise. (3.6.1.9) The measurement to determine compliance with the requirement of mechanically induced antenna noise shall be made in accordance with paragraph 9.1 of SM-A-588564.

4.7.2 Primary Processor. (3.6.2)

The following measurements shall be made in accordance with SM-A-588553:

Scale factor @ 2.5 MHz (3.6.2.2.1)
Scale factor @ 2.38 MHz (3.6.2.2.2)
Scale factor @ 2.62 MHz (3.6.2.2.3)
Slewing Time; 2.50 to 2.62 MHZ (3.6.2.2.4)
Slewing Time; 2.62 to 2.38 MHz (3.6.2.2.5)
AM rejection @ 2.38 MHz (3.6.2.2.6)
AM rejection @ 2.62 MHz (3.6.2.2.7)

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Output Noise with Stable Input (3.6.2.2.8)
Output Bias (3.6.2.2.9)
Filter Response to Step Function (3.6.2.2.10)
Output, with muting (3.6.2.2.11)
Operating Currents (3.6.2.2.12)

4.7.3 Secondary Processor (3.6.3)

The following measurement shall be made in accordance with SM-A-588554:

Initial turn-voltage (3.6.3.2.1)
Settling time (3.6.3.2.2)
Settled Voltage (3.6.3.2.3)
Alarm threshold (3.6.3.2.4)
Log Compressor bias (3.6.3.2.5)
Log compressor output (3.6.3.2.6)
Secondary processor output (3.6.3.2.7)
Repetitive Alarm (3.6.3.2.8)
Mute Output (3.6.3.2.9)
De-logarithmic compression (3.6.3.2.10)
Per-cycle limit (3.6.3.2.11)
Noise Memory (3.6.3.2.12)
Operating Currents (3.6.3.2.13)

4.7.3.1 Secondary Processor Functional Test. (3.6.3.3). This test shall be conducted in accordance with paragraph 6.0 of SM-A-588563.

4.7.4 Sensor Oscillator. (3.6.4)

The following measurements shall be conducted in accordance with SM-A-588555:

Current @ +2.8 VDC (3.6.4.2.1)
RF Power; high frequency (3.6.4.2.2)
RF Power; low frequency (3.6.4.2.3)
Difference frequency, amplitude (3.6.4.2.4)
High frequency, spacial phase, crossover (3.6.4.2.5.1)
A (3.6.4.2.5.2)
DMIN (3.6.4.2.5.3)
Sensor/Oscillator noise (3.6.4.2.6)
Temperature stability (3.6.4.2.7)

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4.7.5 Supplemental circuits. (3.6.5). The following measurements shall be conducted in accordance with SM-A-588557:

+2.8 VDC; load regulation (3.6.5.2.1)
+2.8 VDC; line regulation (3.6.5.2.2)
-2.8 VDC; load regulation (3.6.5.2.3)
-2.8 VDC; line regulation (3.6.5.2.4)
Noise voltage (3.6.5.2.5)
Day/Night Switch Level (3.6.5.2.6)
Disable Pulse (3.6.5.2.7)
Low Voltage disable Point (3.6.5.2.8)
Input Current; armed condition (3.6.5.2.9)

4.7.6 Antenna. (3.6.6) The following measurements shall be conducted in accordance with SM-A-588558:

Input VSWR @ data transmitter frequency (3.6.6.1.1)
Length (3.6.6.1.2)

4.7.7 Antenna coupler. (3.6.7) The following shall be measured in accordance with SM-A-588556:

Loss (3.6.7.2.1)
Input Impedance-Magnitude (3.6.7.2.2)
Input Impedance-Phase (3.6.7.2.3)

4.8 Service condition tests. (3.7)

Pre-Test Measurements as follows shall be made in accordance with SM-A-588564:

Proof Test A

On/Off and disable functions
Sensor oscillator power and frequency
Primary processor and secondary processor outputs
Day/Night Switching levels
Close target rejection
Difference frequency deviation sensitivity
Low voltage

Proof Test B

Same as Proof Test A less On/Off Disable functions.

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Proof Test C

Mechanically induced antenna noise
Detection of Walking Man

For the purpose of the below cited test procedures, the above measurements shall be made at that point calling for pre-test data, in accordance with 3.2.2 of MIL-STD-810.

4.8.1 Temperature (3.7.1)

4.8.1.1 Low temperature. (3.7.1) The equipment minus the antenna shall be subjected to Test Method 502, Procedure I of MIL-STD-810. The storage temperature (step 2) shall be -70°F and maintained at that temperature for 4 hours. The low operating temperature (step 4) shall be +33°F. At steps (5) and (7), test shall be conducted in accordance with SM-A-588564.

4.8.1.2 High temperature. (3.7.1.2) The equipment minus the antenna shall be subjected to Test Method 501, Procedure II of MIL-STD-810. At steps (5), (8), and (11), perform the above measurements. Test shall be conducted in accordance with SM-A-588564. For Step (4), the chamber shall be changed to 74°C (165°F). High operating temperature (Step 7) shall be 64°C (148°F), which includes the effects of solar radiation.

4.8.2 Humidity test. (3.7.2) The equipment, minus the antenna, shall be subjected to Test Method 507, Procedure III of MIL-STD-810. Measurements shall be made during the last 5 hours of each cycle (step 6) and as otherwise specified in Procedure III. At steps (4), (6) and (11), perform the above measurements. Tests shall be conducted in accordance with SM-A-588564.

4.8.3 Altitude Test. (3.7.3) The equipment, minus the antenna, shall be subjected to test method 500, Procedure I of MIL-STD-810. At steps (3) and (5) perform the above measurements in accordance with proof test A of SM-A-588564.

4.8.4 Salt fog test. (3.7.4) The equipment shall be subjected to Test Method 509, Procedure I of MIL-STD-810. The above measurements shall be made after the completion of this test in accordance with proof test A of SM-A-588564.

4.8.5 Sand and dust test. (3.7.5) The equipment shall be subjected to Test Method 510, Procedure I of MIL-STD-810. At step (5), perform the above measurements in accordance with proof test A and C of SM-A-588564.

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4.8.6 Vibration. (3.7.6) The equipment, minus the antenna, shall be subjected to Test Method 514, Procedure IX, Patt I of MIL-STD-810.

4.8.7 Bounce; loose cargo. (3.7.7) The equipment, minus the antenna, shall be subjected to Test Method 514, Procedure XI, Part II of MIL-STD-810. The antenna shall be placed, disassembled, in a separate compartment. Perform the above measurements at the completion of test. Test shall be conducted in accordance with SM-A-588564.

4.8.8 Shock, bench handling. (3.7.8) The equipment, minus the antenna, shall be subjected to Test Method 516, Procedure V of MIL-STD-810.

4.8.9 Shock;drop. (3.7.9) The equipment, minus the antenna, shall be subjected to Test Method 516, Procedure II of MIL-STD-810. Perform the above measurements at the completion of test. Test shall be conducted in accordance with SM-A-588564.

4.8.10 Immersion. (3.7.10) The equipment shall be subjected to Test Method 512, Procedure I of MIL-STD-810 except the depth shall be 15 feet of salt water for 24 hours. Perform the above measurements at the conclusion of this test. Test shall be conducted in accordance with SM-A-588564.

4.8.11 Fungus test. (3.7.11) The equipment shall be subjected to Test Method 508, Procedure I of MIL-STD-810. At the conclusion of the test, the equipment shall be operated so as to perform the above measurements in accordance with proof tests A and C of SM-A-588564.

4.8.12 Wind. (3.7.12) The equipment as set up for operation shall be subjected to forces which simulate wind blowing horizontally at 50 mph, and gusts to 70 mph. The antenna shall be subjected to this test for 20 minutes. At the conclusion of this test, there shall be no evidence of damage and the equipment shall be operated so as to perform the above measurements in accordance with Proof Tests A and C of SM-A-588564.

4.9 Visual and mechanical inspection. (3.8) Visual and mechanical inspection shall be in accordance with MIL-STD-252 and the following inspection: parts, mechanical construction, loose nuts and screws, application of specified treatments and miscellaneous defects. The equipment and controls shall be inspected for proper mechanical and electrical operation. Wiring, soldered connection, grounds, welds, finishes, etc., shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment.

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4.10 Interchangeability demonstration. (3.9) The contractor shall demonstrate that electrical and mechanical interchangeability exists by replacement of each module by other modules of the same type, selected at random from the samples under test. The AN/GSQ-160 shall meet all requirements of this description regardless of which modules are replaced. Test shall be conducted in accordance with SM-A-588563 for Group B inspection or SM-A-588564 for Group C inspection.

4.11 Air seal test. (3.10) The AN/GSQ-160 (with the antenna mounted and battery and transmitter removed) shall be tested in accordance with the procedures of SM-A-588561.

4.12 Electromagnetic interference and compatibility test. (3.11) The contractor shall test the equipment for compliance with the requirements of MIL-STD-461 as defined in 3.11 herein. The test setups and procedures employed shall be in accordance with the requirements of MIL-STD-462. Field tests shall also be conducted to assure compliance with the requirements specified in 3.11 herein.

4.13 Reliability Testing. (3.12) AN/GSQ-160's selected from lots that have passed Group A and Group B tests of this specification, shall be subjected to reliability testing in accordance with MIL-STD-781, applicable paragraphs of MIL-D-55362(EL) and a test procedure approved by the government.

4.13.1 Reliability qualification test. (3.12) The contractor shall perform an attribute type, reliability qualification test as specified below. A sample size of 30 equipments shall be selected at random for reliability testing from first production lot. Each equipment shall be tested for 1080 hours. The test will be successfully passed if one (1) or less relevent failures occur during the test. If two (2) or more relevant failures occur, the equipment will be rejected.

4.13.2 Reliability acceptance test. The contractor shall perform an attribute-type reliability acceptance test as specified below. A sample of 30 equipment shall be selected at random for reliability testing from each group of 500 equipments or portion thereof exclusive of the Reliability Qualification Test Lot. Each equipment shall be tested for 1080 hours. The test will be successfully passed if one (1) or less relevant failures occur during the test. If two (2) or more relevant failures occur, the equipment will be rejected and corrective action taken in accordance with paragraph 4.13.3 below.

4.13.3 Reliability test conditions. The tests described in paragraphs 4.13.1 and 4.13.2 above shall be conducted under the conditions below. The test shall be conducted at a temperature of 100°F ± 5°F. Each equipment

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shall be exercised at least once each 24 hours by simulating field operation and utilization of all operational characteristics. Proper instrumentation shall be provided to insure recognition of failure. A failure is defined as the inability of an equipment to perform its required function. This includes any event which causes departure from performance as required by this specification or the contractor's approved test procedure. The following paragraphs of MIL-STD-781 shall apply:

Failure actions
Verifying repair
Correction action
Preventative maintenance
Restoration of failed equipments
Test Methods

4.13.4 Rework and retest provisions. Once a reject decision has been reached, no rerun can be made of that test on the same or a new sample from the same lot unless positive action consisting of design change, modification, or rework actually related to observed failures has been applied to the lot. Any subsequent rerun must start at zero hours operating time. Acceptance of subsequent lots may be withheld pending corrective action and retest as stated in paragraph 5.7 of MIL-STD-781. Further testing of subsequently produced equipments may be required at the discretion of the procuring activity.

4.14 Acceptance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. Preparation for Delivery.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (See 6.2c).

5.1.1 Level A.

5.1.1.1 Cleaning. Detecting-Transmitting Set AN/GSQ-160 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Detecting-Transmitting Set AN/GSQ-160 shall be dried in accordance with the applicable procedure of MIL-L-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

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5.1.1.4.1 Technical literature. Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Detecting-Transmitting Set, AN/GSQ-160. Each Detecting-Transmitting Set, AN/GSQ-160 shall be packaged method III as follows: Each antenna shall be cushioned by wrapping in paper board conforming to PPP-P-291 type 1, secure wrapping with gummed paper tape conforming to PPP-T-45, grade A, type III. Place the cushioned antenna with the detecting-transmitting set within a close-fitting fiberboard box conforming to PPP-B-636, W5c, center the unit within the box and proceed to cushion the unit within the box with cells or pade or both conforming to PPP-F-320, type CF, variety SW, grade W5c, class weather-resistant designed to protect all projections and absorb the shock of impact in handling and transit. Place the literature packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. Detecting-Transmitting Set, AN/GSQ-160, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents. (See 6.2e).

5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of Detecting-Transmitting Set, AN/GSQ-160, packaged as specified in 5.1, shall be placed in a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, close weather resistant, W5S or V3C. The load shall be "boarded" to the pallet by strapping.

5.2.1.2 Less than palletized load. When quantities per destination are less than a pallet load, the items packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type, PPP-B-621, style 4, class 2, or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds,

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3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto exccept that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.

5.2.2.1 Palletized load. A quantity of Detecting-Transmitting Set AN/GSQ-160's, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized. When quantities per destination are less than a pallet load, a quantity of Detecting-Transmitting Sets AN/GSQ-160, packaged as specified in 5.1 shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet mounted or pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. A quantity of Detecting-Transmitting Set AN/GSQ-160's packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of Detecting-Transmitting Set AN/GSQ-160's, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, repsectively, conforming to PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. Notes.

6.1 Intended Use. Detecting-Transmitting Set AN/GSQ-160( ) is intended for use in forward SEAOPS combat zones as an intrusion detecting device.

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6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Title, number, issue and date of all drawing lists cited in paragraph 2 of this specification.

c. Whether or not the data transmitter modules and/or batteries are to be included in the procurement. (See 3.6.1.8)

d. Serial numbers applicable to the procurement.

e. Level of packaging (Level A or Level C) and level of packing (Level A, Level B, or Level C) required for shipment.

f. When rough handling and functional tests of the packaging are required.

g. Marking and shipping of samples.

h. Place of final inspection.

i. Technical literature required:

TM-11-5895-752-15
Technical Manual for Electromagnetic Detecting-Transmitting Set AN/GSQ-160(U) (TM is not usually packed with each item)
"Installation and Operation", dated August 1971 (Appendix A) (1 each usually packed with each item)

j. Whether the following items are to be supplied:

Data Transmitter T-1243/GSQ
Encoder KY-678/GSQ
Code Plug KY-677/GSQ
Battery BA-1549
Counterpoise SM-D-588223

These items are part of the AN/GSQ-160 but are normally not supplied by the contractor. Limited quantities required by inspection are usually GFE'd to the contractor.

k. Whether TA-386 interface connector is CFE or GFE - Normal requirement 110% includes attrition.

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1. GFP required for inspection and test (usually as follows):

AN/GSQ-46	2 ea
T-1243/GSQ	36 ea (6 - Prod test; 30 Rel. test)
KY-678/GSQ	36 ea (6 - Prod test; 30-Rel. test)
KY-677/GSQ	36 ea (6 - Prod test; 30-Rel. test)
TA-383 connector	25 ea (for inspection test fixtures)
BA-1549/U	12 ea plus as required for Rel. test

m. Models or standard sample.

n. Gages and special test equipment.

6.3 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.5).

6.4 Location of operational inspection and air-seal test. It is desirable that the operational inspection (4.12) and the air-seal test (4.11) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GSQ-160W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

Custodian:
Army-EL

Preparing Activity
Army-EL

Project 5840-A163

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INSTALLATION AND OPERATION

1. SCOPE. This instruction sheet provides installation and operation data for Electromagnetic Detecting-Transmitting Set, AN/GSQ-160. All particulars are detailed in DTM 11-5895-752-15.

2. DESCRIPTION. The AN/GSQ-160 is hand-buried for unattended operation (figure 1). It consists of two-piece whip antenna AN-2542/GSQ-160, upper subassembly MX-8846/GSQ-160, battery box CY-7046/GSQ-160 containing battery BA-1549/U, and four counterpoise wire spools. Subassembly and battery box latch together as an 8×8×6 inch waterproof case weighing 11 lbs. The antenna, 46 inches long (upper section curved for camouflage), threads to an insulated connector. The strap securing the antenna protective cap is used as a carrying handle.

Caution: The AN/GSQ-160 contains a non-hazardous anti-tamper disabling circuit. Switching out of ARM before setting the recovery code switches, or handling or tampering when the ARMING/RECOVERY switch is in ARM will disable the unit. Read paragraphs 3 and 4 before handling.

3. CONTROLS. Three: on top of the case (figure 2).
a. ARMING/RECOVERY Switch. OFF/SAFE for handling and storage; ON/TEST for field test; ARM (pull-turn) for operation with anti-tamper disable active.
b. RECOVERY CODE Switches. One marked 1 through 11 and SET; the other marked A through K and SET. Set to a letter-number combination to bypass anti-tamper circuit and switch out of ARM for handling or recovery without triggering the disable. Every AN/GSQ-160 has a unique recovery code combination: B-9, D-4, etc. Know the recovery code before handling. Direct support levels list recovery codes by unit serial number.
c. Day/Night Sensor. Photocell; covered by antenna protective cap for continuous operation; left uncovered for operation only at night.

4. ANTI-TAMPER DISABLE. With battery installed and ARMING/RECOVERY switch set to ARM, an AN/GSQ-160 will disable if ONE of the following occur:
a. ARMING/RECOVERY switch moved out of ARM before recovery code switches are set to safe combination.
* b. The AN/GSQ-160 is tilted more than 45° ±8.
c. Upper Subassembly and battery box are unlatched.
d. Battery voltage drops below certain minimum levels.
* e. The AN/GSQ-160 is accelerated in any direction for more than 3/10ths of a second.

5. TOOLS AND EQUIPMENT REQUIRED. A Rucksack, MIL-R-43373A, or equivalent; an entrenching tool, pliers, a knife, and Post-Implant Test Device, AN/xxx.

6. PRE-DEPLOYMENT INSPECTION. Verify that ARMING/RECOVERY switch is OFF/SAFE. Then:
a. Verify AN/GSQ-160 is securely latched and has no dents or punctures. Ignore scratches and small dents.
b. Verify antenna insulator is clean, not cracked or abraded, that antenna connector is clean and protective cap secure.
c. Inspect antenna sections and connectors. The lower section must not be dented or bent. Replace if necessary. Tape or strap the two sections together for transport.
d. Verify recovery code combination against direct support serial number code lists. NEVER leave recovery code data attached to AN/GSQ-160.

7. TRANSPORT. Hand-carry by strap handle or sling from shoulder strap. Use rucksack for two or more units. Hand-carry antenna or secure to rucksack as convenient. Secure counterpoises, tools, and AN/xxx to avoid damage or loss.

8. DEPLOYMENT. Select a site with natural foliage for camouflage and no obstructions to case burial.
a. Dig a hole 9 inches square and 6 inches deep. Use antenna insulator as handle to place AN/GSQ-160 in hole with case top at grade level; set case level (figure 3).

Note: Use counterpoises ONLY if soil has NO natural moisture. Connect counterpoise wires to case posts. Bury wires at least two inches deep.

b. Connect upper and lower antenna sections. Retain end plugs. Remove insulator protective cap and place cap on day/night sensor. Thread antenna onto connector tightly (waterproof fitting) aligning antenna curve with prevailing curve of nearby vegetation.
c. Clear a 2-foot area around antenna of foliage which may rub or strike the antenna. DO NOT remove more growth than necessary.
d. Push soil around case flush with top surface and pack firmly. Proceed with operation (paragraph 9).

9. OPERATION. Clip AN/xxx to upper antenna (figure 4) and adjust to view lamp from 10 to 16 ft away. Verify antenna cap covers day/night sensor.
a. Position ARMING/RECOVERY switch to ON/TEST. Move back 10 ft, keeping lamp in view. Remain still for five minutes. There should be no flashes. AN/GSQ-160 stabilization time is about five minutes.
b. Move out 10 ft and return to AN/GSQ-160 looking for AN/xxx flash at 5-to-10 second intervals. Or an assistant can walk 0 to 20 ft while operator observes.
c. If flashes do not occur, repeat steps a and b twice. If flashes are still not observed, initiate recovery (paragraph 10). If flashes are seen, remove and retain AN/xxx.
d. For night-only operation, remove cap from day/night sensor and conceal beside case.
e. Position both code switches to SET, and ARMING/RECOVERY switch to ARM (pull-to-turn).

* Caution: Any tilt, shock, or movement of AN/GSQ-160 in ARM mode may trigger the disable, making unit useless.

f. Cover counterpoises and case with soil. Leave 1/2-inch clearance between antenna insulator bottom and surface soil. Ensure day/night sensor window is not covered (figure 5) for night-only operation.
g. Retain or bury all litter; camouflage site.

10. RECOVERY. Recover AN/GSQ-160 before battery run-down to prevent disablement (DTM 11-5895-752-15).
a. Verify recovery code for AN/GSQ-160 to be retrieved against DS serial number and code lists.
b. Remove soil from top of unit. Set recovery code switches to safe recovery combination. If switch marks are obscured, turn switches fully counterclockwise, then rotate clockwise to code combination by counting positions.

Caution: Failure to set recovery code switches will disable AN/GSQ-160.

c. Set ARMING/RECOVERY switch (pull-turn) OFF/SAFE.
d. Disconnect counterpoise cables and lift case free using antenna insulator as handle.
e. Unthread antenna, separate sections, and secure for transport. Install end plugs if available.
f. Clean inside of antenna protective cap and thread cap onto antenna connector. Counterpoises may be discarded.

NOTE: All references to the tilt disable function refer only to serial numbers 533 and below. Those serial numbered 534 and above do not employ the tilt function.

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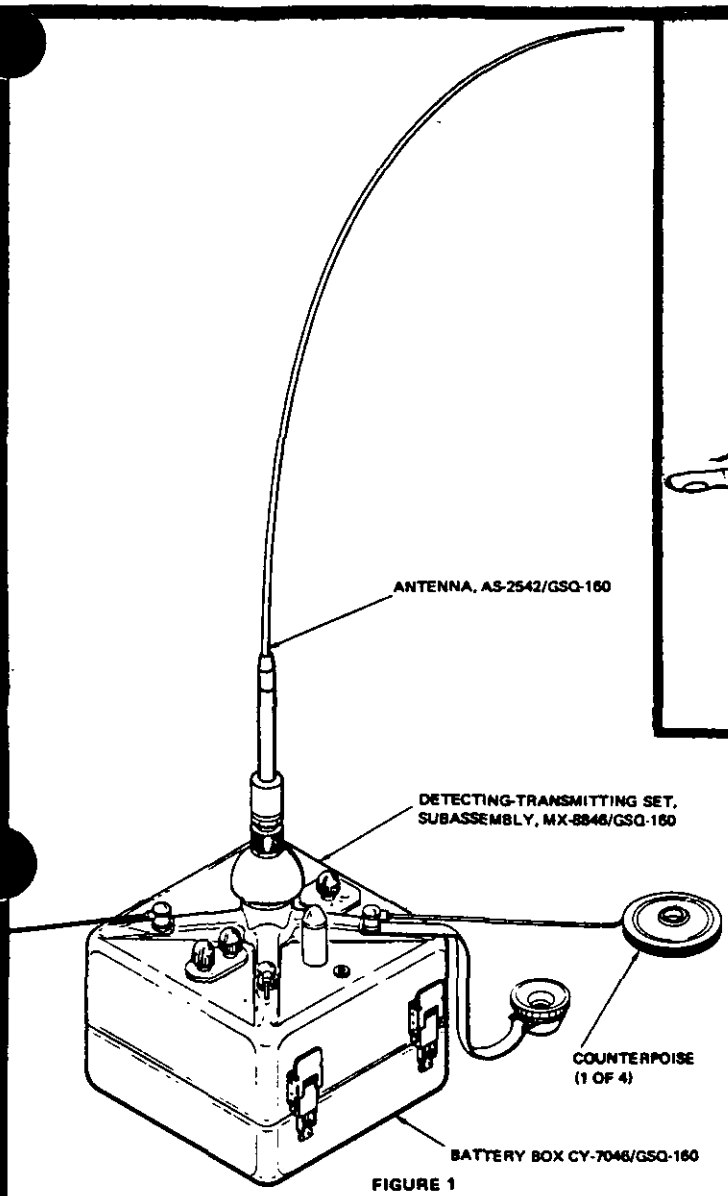


FIGURE 1

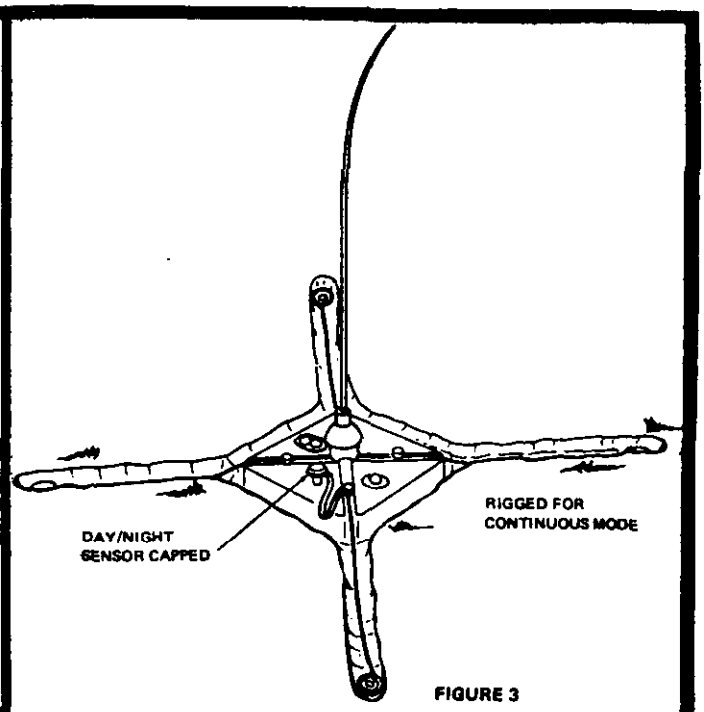


FIGURE 3

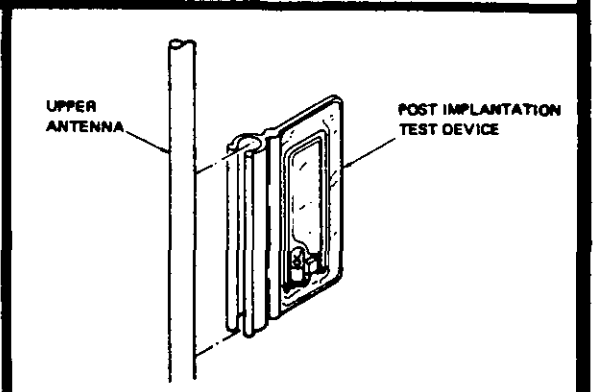


FIGURE 4

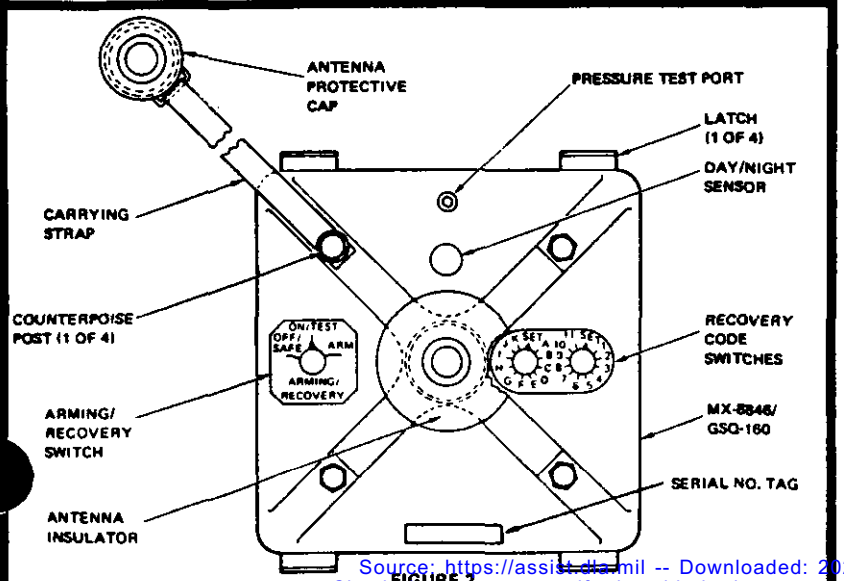


FIGURE 2

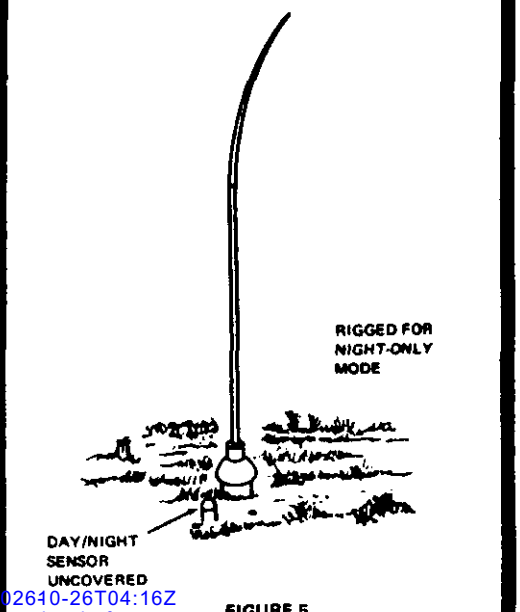


FIGURE 5

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APPENDIX B

DETECTING-TRANSMITTING SET
ELECTROMAGNETIC, AN/GSQ-160

Serial Number _____

Security Code _____

Channel Number _____

Code Assignor _____

Date and Stamp _____

Unit Location _____

_____

_____

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APPENDIX C

and

APPENDIX D

Will Be Added When

General Dynamics Corporation

Provides the MYLAR

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